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A Appendix

A.1 NUPA Test

A.1.1 Representations

We present the four representations as follows:

- **Integer**: we use no comma or point as a *digit group separator* like 1234567. The integer has only one part as itself. In this paper, we have not considered negative numbers for the time being.
- **Float**: A float has two parts: integer and decimal. We use a decimal point to split these two parts and also do not use any digit group separator. An example is 1234.567891. Trailing zeros in the decimal part are usually omitted.
- **Fraction**: A fraction has two parts: numerator and denominator and we use a "/" to separate the numerator and denominator parts. Unless otherwise specified, all fractions mentioned in this paper are in their simplest form (that is the numerator and denominator are coprime),